

Army re_entry — BDP + 3x3 data story (AWS port)

Purpose: Uniform 9-panel porch deck for Army talent data — same layout as MBB, tenure, football, and Legends.

Status: v0.3 — panels 2–9 live (Act II CCT + elite pond scaled from tenure/MBB).

AWS folder layout (520 root = e.g. `Network_1P_shell`)

```
Network_1P_shell/          ← Jupyter cwd (520 notebook here)
├─ 520_pipeline_cox_working.ipynb
├─ pipeline_config.py
├─ big_dfs/
│   └─ df_pipeline_11_cox_analysis.feather  ← Cell 11 input
├─ talent/
│   └─ re_entry/                ← this folder
├─ sports/
│   └─ scripts/                 ← compositor (2 files required)
│       ├── build_data_story_mosaic.py
│       └─ story_page_layout.py
```

No `PYTHONPATH` needed — scripts bootstrap paths automatically when run from 520 root.

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Path	Required
<code>talent/re_entry/*</code> (all <code>.py</code> , <code>.sh</code> , <code>manifests/</code> , <code>output/.gitkeep</code>)	Yes
<code>sports/scripts/build_data_story_mosaic.py</code>	Yes
<code>sports/scripts/story_page_layout.py</code>	Yes
<code>sports/scripts/hero_plot_style.py</code>	Yes (BDP + HERO panels)
<code>sports/scripts/empirical_team_interval_overlap.py</code>	Yes (panel 5 overlap)
<code>sports/scripts/gallery_mathtext.py</code>	Yes (panel 5 mathtext)
<code>sports/scripts/plot_provenance.py</code>	Yes (Act II bin badges)
<code>sports/541_grandchild_homophily_assign.py</code>	Optional (H_sort in overlap title)

Mosaic compositor only imports `story_page_layout.py` — not `hero_plot_style` .

Run (from 520 root)

One shot:

```
bash talent/re_entry/run_army_bdp_pipeline.sh
```

Or step by step:

```
./talent/re_entry/army_basic_plots.py --all
./talent/re_entry/army_act2_probes.py --plot all_probes
./talent/re_entry/army_hero_slide_plot.py
./talent/re_entry/build_army_data_story.py
```

Outputs:

- `talent/re_entry/output/basic_data_plots/ARMY_BDP_*.png`
- `talent/re_entry/output/act2/ARMY_CCT_*.png` · `ARMY_ELITE_*.png`
- `talent/re_entry/output/hero/ARMY_HERO_ew8_z_pool_minus_mean_snr_fwd_run1.png`
- `talent/re_entry/output/data_story/ARMY_DATA_STORY_run1_3x3.png`

Prerequisites

- 520 through Cell 11 (`pipeline_config_17_1` or your Run 1 config).
- Feather at `./big_dfs/df_pipeline_11_cox_analysis.feather` (via `load_feather` / `store_feather`).
- Panel 5:** `pipeline_config.py` must include `snr_rater_bwd` in `base_time_varying_cols` so Cell 11 exports the senior-rater pool key (see `AWS_UPLOAD_CHECKLIST.md`).

HERO defaults (Cell 11 CR CIF-bar aligned)

Edit one line in `army_hero_slide_plot.py`:

```
plot_var = "z_pool_minus_mean_snr_fwd"
```

Defaults: **EW8**, min pool **3**, zero-OER + NaN filters, **last-event** promotion coding.

Porch-deck mode: `./talent/re_entry/army_hero_slide_plot.py --outcome ever_promoted --n-bins 12`

9-panel standard

#	Panel	Army v0
1	Cohort (text)	auto-refresh from BDP manifest
2	$\hat{A}_i$ and T_j	<code>ARMY_BDP_Ai_Tj_run1.png</code>
3	LOO pool distribution	<code>ARMY_BDP_pool_minus_mean_loo_run1.png</code>
4	Promotion mass vs $\hat{A}$	<code>ARMY_BDP_promotion_mass_ecdf_run1.png</code>
5	Pool interval overlap	<code>ARMY_BDP_pool_interval_overlap_run1.png</code>
6	Pool size $ T_j $	<code>ARMY_BDP_pool_size_run1.png</code>
7	CCT (fixed $\hat{A}$ $z \in [1,2]$, Q8 LOO)	<code>ARMY_CCT_promotion_rate_pool_loo_run1_z1_2_q8.png</code>
8	Elite pond (top 20% $\hat{A}$, PW 3+5 LOO)	<code>ARMY_ELITE_pond_loo_pw3p5_run1_top20.png</code>
9	HERO	<code>ARMY_HERO_ew8_z_pool_minus_mean_snr_fwd_run1.png</code>

Compositor: `sports/scripts/build_data_story_mosaic.py` (shared with MBB/tenure).

Updated 2026-09-18 — *path bootstrap, big_dfs, CR-aligned HERO.*